Routing Legibility in Cooperative MARL: Measuring the Translation Gap Between Theory-Informed Roles and Learned Conventions

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Abstract

Classical game theory suggests which strategic dependencies should matter; cooperative multi-agent reinforcement learning instead discovers conventions through non-stationary learning dynamics. We study the gap between those two through routing legibility: whether a trained policy exposes clear, measurable role-to-role dependency structure. We introduce a compact analysis framework built around three measurements: a role-routing matrix, a formation-sensitivity metric ($\Delta_{\max}$) for models that contain ally self-attention, and input-gradient attribution for MLP baselines. We evaluate the framework across two domains: a custom grid-world cooperative task (Domain 1, three-role MiniGrid) and a custom SMACv2 scenario (Domain 2, three-role Terran micromanagement). Both domains use the same five controlled policy conditions under a fixed MAPPO training setup; Domain 2 additionally includes a 3v3–9v9 scaling study for the two central conditions. The two-domain design lets us separate architectural effects on routing legibility from any single environment's dynamics. The main result is structural: when role labels are present in both the self token and ally tokens, attention-based policies produce substantially more concentrated role-conditioned routing than MLP baselines. The full attention model yields a routing-matrix standard deviation of 0.275, versus 0.055 for the MLP pseudo-routing matrix, and removing role labels from a shared attention policy reduces win rate by 17.8 percentage points (0.652 to 0.474). In scaling, the full attention model has a large 3v3 advantage over Role+MLP (last-10 win rate 0.817 vs. 0.520), while the advantage narrows in 6v6 and 9v9, exposing larger-team optimization instability. Three follow-up analyses sharpen the structural reading: (i) the role-conditioned routing structure is preserved at 9v9 even when the win-rate gap narrows; (ii) a 3v3-trained attention checkpoint zero-shots to 9v9 at 0.213 win rate, exceeding the from-scratch 9v9 attention baseline (~ 0.15); (iii) a padding-content ablation bounds the padded-slot confound at $\leq 15\%$ of the 3v3 gap, leaving the architectural residual at $\geq 85\%$. Two theory-informed behavioral predictions did not hold in the expected direction. Support, not Vanguard, had the highest $\Delta_{\max}$, and Strike routed primarily to Vanguard rather than Support. We treat this not as a failure of the mechanism, but as the paper's main scientific observation: architecture makes strategic dependency legible, while learning dynamics determine which convention is selected. The contribution for the NExT-Game workshop is a compact diagnostic for studying how learned strategic conventions depart from theory-informed prescriptions.

1. Introduction

As AI systems move from isolated prediction to strategic interaction, one of the central questions for both game theory and machine learning is no longer only what equilibrium should exist, but what convention a learning system will actually settle on. Classical game theory provides role-sensitive prescriptions under explicit assumptions about information, incentives, and rational response. Cooperative MARL agents, by contrast, learn through decentralized gradient updates in non-stationary environments. Even when role structure is known, the resulting coordination pattern need not resemble the theory-informed prescription. This mismatch is the translation gap we study.

Our focus is not performance alone. We ask whether architecture affects the legibility of learned strategic

dependency: can we read off, from the policy itself, which agent roles condition on which other roles? In heterogeneous teams, this is a routing problem. A Strike agent may need to anchor on a Vanguard, a Support, or neither, depending on the convention learning has discovered. If those dependencies are encoded diffusely inside a flat MLP, they may be difficult to measure. If they are encoded through role-conditioned attention, they may become directly inspectable.

We make a scoped mechanism claim. We do not claim that attention guarantees the "correct" game-theoretic convention, nor that MLPs cannot use role labels at all. We claim that placing role labels in both self_tok and ally_tok creates a direct architectural pathway for role-pair routing in attention, and that this pathway makes emergent strategic dependency more measurable than in flat MLP baselines.

We evaluate this claim on two cooperative MARL domains chosen for their complementary dynamics: Domain 1, a custom three-role grid-world built on MiniGrid that isolates routing legibility under simple, fully-observable cooperation; and Domain 2, a custom SMACv2 scenario using three Terran unit types (Marine, Marauder, Medivac) as role proxies, which exercises the same architectural primitives in a richer continuous-control setting and adds a 3v3–9v9 scaling study. Both domains use the same five ablation conditions, the same MAPPO training setup, and the same routing-analysis pipeline; agreement across domains is what licenses our structural claim. The paper's contribution is not state-of-the-art performance; it is an analysis framework — the role-routing matrix, the formation-sensitivity metric $\Delta_{\max}$, and gradient attribution comparison — that makes routing legible and comparable across architectures. This framework reveals:

- Attention produces much more structured role-conditioned routing than MLP.
- Removing role labels from a shared-policy attention system substantially reduces performance.
- The specific convention selected by learning differs from our theory-informed behavioral prediction in a stable, tactically coherent way.

That third result is especially important for the NExT-Game workshop framing. The paper is not just about attention as an inductive bias; it is about how strategic conventions emerge under learning dynamics even when theory suggests a different routing profile. The analysis framework is our main contribution because it makes that divergence measurable.

2. Related Work

Role discovery in cooperative MARL. ROMA (Wang et al., 2020) learns latent role representations automatically from trajectories using variational inference, enabling emergent specialization without pre-defined role semantics. Our work takes the complementary view: given pre-defined role labels, we analyze the mechanism by which different policy architectures exploit those labels. This is a conditioning problem, not a discovery problem. The question is not "what roles emerge?" but "does the architecture make role labels structurally load-bearing for entity selection?"

Relational reasoning in reinforcement learning. Zambaldi et al. (Zambaldi et al., 2018) introduced self-attention as a relational inductive bias for single-agent reinforcement learning, demonstrating that agents can learn to attend to task-relevant entities in Atari environments. We extend this to the heterogeneous multi-agent setting, where the relevant entity is not the spatially proximate one but the role-compatible one. Entity-centric observation representations (Christianos et al., 2020) provide the structured input format that makes this extension natural: placing role labels in distinct positions within self_tok and ally_tok creates the structural conditions for role-conditioned Q/K routing that is absent in flat observation baselines.

Cooperative optimization without role interpretability. QMIX (Rashid et al., 2018) uses centralized monotonic value mixing to enable cooperative optimization with decentralized execution. MAPPO (Yu et al., 2022) extends PPO to the cooperative multi-agent setting with a centralized critic. Neither framework is designed to produce interpretable role specialization: the network can learn role-differentiated behavior incidentally, but there is no structural path for role-pair-conditioned entity selection. We use MAPPO as the training algorithm across all five conditions and demonstrate that the routing structure is an architectural property (attention vs. MLP), not an optimization property (MAPPO is constant across conditions).

Attention-based MARL. MADDPG (Lowe et al., 2017) and subsequent work (Iqbal & Sha, 2019) applied attention to multi-agent communication and value decomposition. These works use attention for agent-to-agent communication or value aggregation, not for structured role-conditioned entity selection within a single agent's observation. Our cross-attention mechanism differs in that the query is the agent's self-embedding (role-conditioned) and the keys/values are ally embeddings (also role-

conditioned), making the routing computation explicit and attributable via the routing matrix analysis.

3. Theory: Role-Label-Conditioned Routing

3.1. The Routing Problem in Heterogeneous MARL

In cooperative MARL with heterogeneous roles, agents must not only follow their own behavioral mandate but also identify which teammates are most relevant to their decision at each timestep. This is a routing problem: given a set of ally entities, which subset should a given agent condition on, and how strongly?

Role labels provide a natural signal for this selection. A ranged attacker (Strike) benefits most from knowing the position of the frontline tank (Vanguard), not the healer. A healer (Support) prioritizes the most at-risk ally. These routing preferences are determined by role-pair relationships, not spatial proximity alone. The central question of this paper is: do different policy architectures differ in how effectively they exploit role labels as routing signals?

3.2. MLP Cannot Structurally Encode Role-to-Role Routing

In an MLP policy, the role label is concatenated with all other observation features and processed by shared weight matrices:

$$h = \text{MLP}([o_{\text{pos}},\ o_{\text{hp}},\ r_{\text{onehot}},\ \ldots])$$

The computation "which entity to condition on given my role" is not structurally encoded. It must be implicitly learned as a feature interaction buried in weight products $W_1 W_2 \cdots W_n$. The MLP has no structural inductive bias toward routing by role-pair. Gradient attribution confirms that role-label signal is much weaker than spatial signal in the MLP baseline (Section 5), while the attention mechanism produces markedly more structured routing patterns.

Note that this does not claim MLP routing is structurally impossible. The claim is calibrated: attention provides a more direct structural inductive bias for role-pair routing. Whether this bias activates is verified empirically.

3.3. Why Self-Attention Provides a Direct Routing Path

In the proposed architecture, role labels appear in two positions in the observation:

- self_tok[4:7]: the agent's own role one-hot, encoded into the Query.
- ally_tok[12:15], ally_tok[20:23]: each ally's role one-hot, encoded into the corresponding Key/Value.

The cross-attention score between agent i (role r_i) and ally j (role r_j) is:

$$\text{attn}_{ij} = \text{softmax}\left(\frac{(W_Q \cdot \mathbf{x}_i)(W_K \cdot \mathbf{x}_j)^\top}{\sqrt{d_k}}\right)$$

where $\mathbf{x}_i$ includes r_i and $\mathbf{x}_j$ includes r_j. Because role labels condition both Q and K simultaneously, the attention score is a direct function of the role-pair (r_i, r_j), not just spatial features. The network can learn role-to-role routing via weight matrices W_Q and W_K that project the role one-hot into the query and key spaces respectively.

This computation is short (one matrix multiplication per head) and attributable: the routing can be read out directly from attention weights, as done in Section 5.

3.4. Intra-Set Self-Attention: Relational Context Before Routing

Before cross-attention, each entity group passes through an Intra-Set Self-Attention block. This provides relational context within the ally set before the agent's query is formed. Specifically, the ally token for Vanguard can be enriched by its relationship to the Support token before entering the cross-attention Key computation.

This explains the performance gap between Condition 1 (Full) and Condition 2 (No Self-Attn): without intra-set contextualization, Keys and Values entering cross-attention are computed from raw entity embeddings without relational structure. Intra-set self-attention contributes beyond the role-routing path, but the routing mechanism itself is primarily located in cross-attention.

3.5. Testable Predictions

The structural argument generates four testable predictions, evaluated in Section 5:

1. Attention produces more structured role routing than MLP (routing matrix std is higher). [PASS]
2. Removing role labels degrades performance by >5pp when a shared policy is used. [PASS]
3. Formation sensitivity ($\Delta_{\max}$) is highest for Vanguard (frontline, formation-sensitive). [REVISED: Support highest]

4. Strike attends preferentially to Support (healer-seeking behavior). [REVISED: Strike→Vanguard dominant]

Predictions 3 and 4 are revised: the emergent routing pattern differs from the domain-knowledge prediction but remains tactically coherent. This is the paper's central translation-gap result rather than a failure case.

4. Method

4.1. Environment

We evaluate on a custom SMACv2 scenario using three Terran unit types as heterogeneous role proxies: Marine (Strike, ranged DPS), Marauder (Vanguard, melee frontline), and Medivac (Support, healer). The team of three agents operates against opponents on a generation map with 10 total agents per side. Episodes terminate at the environment's default step limit; win rate is the primary metric.

SMACv2 does not natively support pre-defined role labels in observations. We inject role identity as a 3-dimensional unit-type one-hot into both the agent's own token and each ally token, preserving the structural condition required for role-conditioned Query/Key routing (Section 3).

4.2. Entity-Centric Observation

Each agent receives a 65-dimensional entity-centric observation structured as four named token groups:

- self_tok $\in \mathbb{R}^8$: position(2), health(1), shield(1), role_onehot(3), pad(1).
- ally_tok $\in \mathbb{R}^{16}$: 2 allies × 8 dims with rel_pos(2), health(1), shield(1), role_onehot(3), alive(1).
- enemy_tok $\in \mathbb{R}^{16}$: 2 enemies × 8 dims with rel_pos(2), health(1), shield(1), type_conf(3), alive(1).
- zone_tok $\in \mathbb{R}^{18}$: 3 zones × 6 dims with rel_pos(2), owned(1), contest(1), value(1), pad(1).
- masks $\in \mathbb{R}^7$: validity flags for ally, enemy, and zone slots.

Role one-hots occupy 3 dimensions in self_tok[4:7] and in each ally slot (ally_tok[12:15], ally_tok[20:23]). This placement is structural: the attention mechanism encodes these into Q (from self) and K/V (from ally), creating a direct role-pair routing path (Section 3).

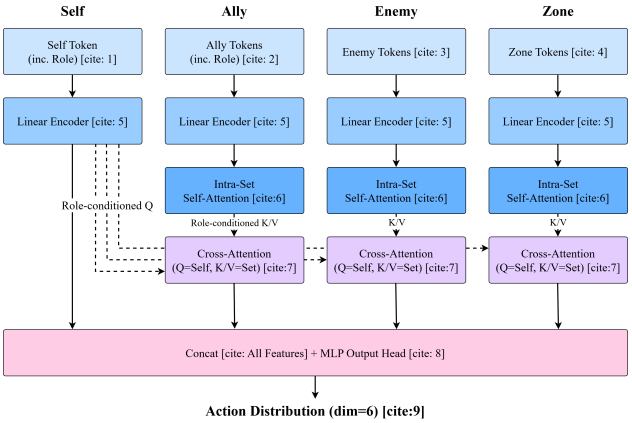


Figure 1. SLIC architecture (Condition 1 — Full). Four entity token groups are encoded by dedicated linear layers. Intra-Set Self-Attention contextualises each entity set independently. Cross-Attention uses the self-embedding as query and each set as key/value. Red arrows highlight the role-conditioned routing path: the role one-hot in self_tok is projected into Q, while the role one-hots in ally_tok are projected into K/V, enabling structured role-pair attention routing without architectural hand-coding.

4.3. Policy Architecture

We implement five ablation conditions as swappable policy classes sharing the same interface. All take the 65-dim observation and produce a 6-dimensional action distribution. Conditions 1, 2, 3, and 5 use three independent policies (one per role); Condition 4 uses a single shared policy for all three roles.

Condition 1 — Full (proposed). Encodes each entity group with a dedicated linear layer (dim 64), applies Intra-Set Self-Attention to each set (2-head MultiheadAttention with residual and LayerNorm), then applies Cross-Attention with the self-embedding as query and each set as key/value. The output MLP concatenates all three context vectors plus the self-embedding:

$$\pi = \text{MLP}([\mathbf{c}_{\text{ally}}, \mathbf{c}_{\text{enemy}}, \mathbf{c}_{\text{zone}}, \mathbf{e}_{\text{self}}]) \in \mathbb{R}^{128 \to \text{action_dim}}$$

Condition 2 — No Self-Attn. Removes the three Intra-Set Self-Attention blocks. Encoded ally/enemy/zone tensors feed directly into Cross-Attention, eliminating intra-set relational structure while retaining cross-set routing.

Condition 3 — Role+MLP. Removes all attention. A flat MLP processes the full 65-dim observation: $\text{Linear}(65, 256) \to \text{ReLU} \to \text{Linear}(256, 128) \to \text{ReLU} \to \text{Linear}(128, \text{action_dim})$. Role one-hots are present in the input vector but processed identically to all other features.

Condition 4 — No-RoleLabel (shared policy). Same architecture as Condition 1, but role one-hot dimensions are zeroed in both self_tok[4:7] and all ally_tok role dims before encoding. A single shared policy handles all three roles; role-specific rewards still apply during training. This is the meaningful isolation of label contribution: with a shared policy, the network cannot specialize by trajectory distribution alone.

Condition 5 — Flat-MLP baseline. Same as Condition 3 with role one-hots zeroed. No labels, no attention. This is the performance floor.

Padding safety. All attention computations apply key_padding_mask from the observation masks. A safety function force-unmasks slot 0 if all slots are masked, preventing NaN in softmax.

4.4. Training Setup

All conditions use MAPPO (Yu et al., 2022) with the following hyperparameters, kept constant across all conditions:

Hyperparameter	Value
Algorithm	MAPPO
Total steps	2M per seed
Seeds	3
GAE λ	0.95
Discount γ	0.99
PPO clip ε	0.2
Entropy coefficient	0.01
Optimizer	Adam, lr $= 3 \times 10^{-4}$
Batch size	512
Mini-batch size	64
PPO epochs per update	4
Centralized critic	MLP on concatenated observations ($65 \times 3 = 195$ dims)
Checkpoint interval	every 200k steps

Independent policies (Conditions 1–3, 5) are updated only on trajectories from their respective role. The shared policy (Condition 4) is updated on all trajectories.

4.5. Reward Structure

Role-specific rewards encode the behavioral mandate of each unit type:

- Strike (Marine): Approach bonus toward enemies + attack bonus + too-close penalty (range enforcement).
- Vanguard (Marauder): Approach bonus + melee engagement bonus when in close combat range.
- Support (Medivac): Approach-to-lowest-HP-ally bonus + heal event bonus + rear-arc positioning bonus.

Team reward is mixed with individual reward: $r_{\text{final}} = 0.8 \cdot r_{\text{individual}} + 0.2 \cdot \bar{r}_{\text{team}}$.

5. Experiments

5.1. Two-Domain Experimental Setup

We evaluate the same five ablation conditions across two cooperative MARL domains that share an identical entity-centric observation schema and the same MAPPO training setup. Each condition is trained for 2M environment steps in each domain. All conditions in a given domain share identical hyperparameters from a single configuration file. Reporting two domains, rather than treating one as primary and the other as a replication, is intentional: an architectural claim about routing legibility should hold across more than one environment family before it is treated as structural.

Domain 1 — Three-role grid-world (MiniGrid). A custom 15×15 grid built on MiniGrid with three role-specialized agents (Strike, Vanguard, Support) cooperating to capture and hold zones under role-specific reward shaping. Observations are entity-centric (self_tok, ally_tok, enemy_tok, zone_tok) with role one-hots embedded in both self_tok and each ally slot. Domain 1 isolates the routing-legibility question in a simple, fully observable, low-dimensional setting where confounds from continuous control and stochastic combat outcomes are minimal.

Domain 2 — SMACv2 micro-management. A custom SMACv2 scenario using three Terran unit types as role proxies: Marine (Strike), Marauder (Vanguard), and Medivac (Support). The 65-dimensional entity-centric observation in the 3v3 setting (and 182-dimensional padded form for the 3v3–9v9 scaling study) contains the same four named token groups: self_tok (8 dims: position, health, shield, role one-hot), ally_tok (16 dims: 2 allies × 8 in 3v3), enemy_tok (16 dims), and zone_tok (18 dims), followed by validity masks. Role one-hots occupy 3 dimensions within both self_tok and each ally slot in ally_tok; this placement is load-bearing for the routing mechanism (Section 3). Domain 2 stresses the same primitives in a richer continuous-control regime and additionally enables the scaling study (Section 5.3).

The shared schema across the two domains lets us run an identical analysis pipeline (Track 2) on both, so any

architectural pattern that appears can be directly compared rather than inferred across mismatched designs. The remainder of this section reports Domain 2 in detail (where the scaling study lives), then Domain 1 as a co-equal cross-domain check.

5.2. Domain 2 (SMACv2) — Track 1: Performance Comparison

Table 1 reports the final win rate for all five conditions.

Table 1. Track 1 performance on SMACv2. Win rate is the mean of final evaluation checkpoints.

Condition	Win Rate	Std
Cond 1: Full	0.652	0.079
Cond 2: No Self-Attn	0.551	0.210
Cond 3: Role+MLP	0.440	0.143
Cond 4: No Role Label (shared)	0.474	0.204
Cond 5: Flat MLP	0.494	0.064

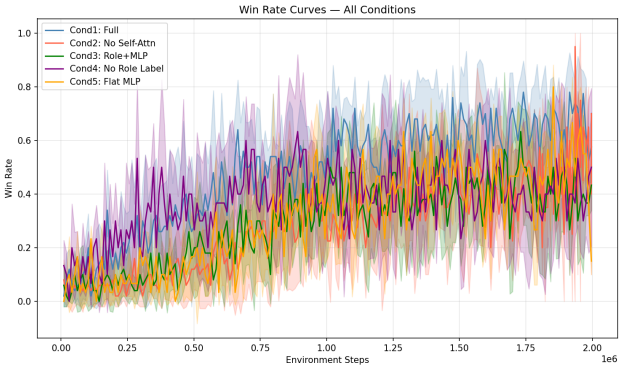


Figure 2. Learning curves for all five conditions over 2M training steps. The full attention condition converges highest, while removing intra-set self-attention or removing role labels lowers performance.

Three observations matter for the paper's core claim. First, attention helps: Condition 1 outperforms Condition 3 by 21.2 percentage points (0.652 vs. 0.440). Second, intra-set context helps: Condition 2 trails Condition 1 by 10.1 points, indicating that contextualizing ally entities before routing adds value. Third, labels matter for attention: Condition 4 is 17.8 points below Condition 1, supporting the claim that explicit role labels are structurally useful in the attention architecture.

Notably, Condition 5 slightly outperforms Condition 3 (0.494 vs. 0.440). This does not weaken the main thesis; it sharpens the narrower interpretation that adding role labels to an MLP is not enough to produce useful role-conditioned routing.

5.3. Domain 2 — Scaling Study: 3v3, 6v6, and 9v9

We additionally evaluate whether the contrast between the full attention model and the Role+MLP baseline survives larger team sizes. We train balanced 3v3, 6v6, and 9v9 variants with the same role proportions: Strike, Vanguard, and Support appear in equal counts on the controlled team. The enemy composition scales in parallel while preserving the original asymmetric design: enemies contain Strike and Vanguard proxies but no Support unit. All scaling runs use the same MAPPO implementation, a true centralized critic over concatenated global observations, and identical hyperparameters within the scaling study.

Table 2 reports two less volatile metrics than a single final checkpoint: the mean win rate over the last ten evaluations, and the mean win rate across training as an AUC proxy. For 9v9 Condition 1, seed 2 was interrupted before completion and is excluded; seed 3 is used as the third completed seed.

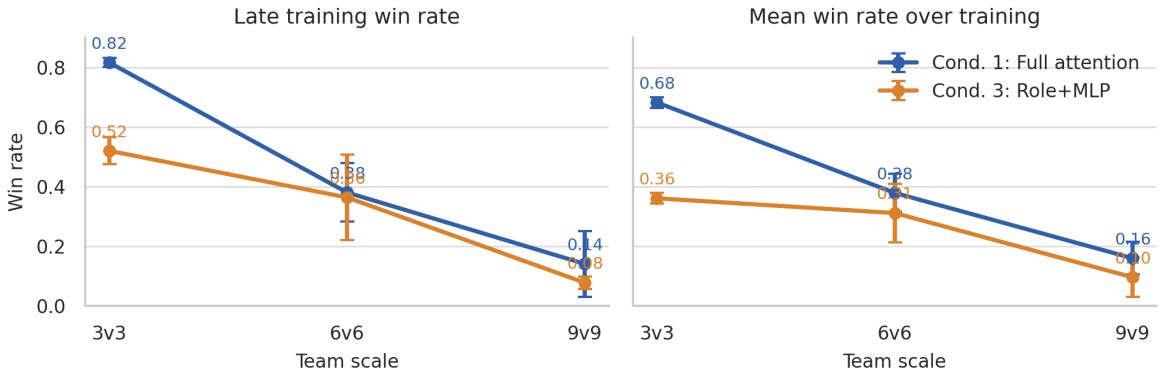
Table 2. Scaling results for Condition 1 (Full attention) and Condition 3 (Role+MLP). Values are mean ± std over three completed seeds.

Scale	C1 Last-10	C3 Last-10	C1 AUC	C3 AUC
3v3	$\mathbf{0.817 \pm 0.015}$	0.520 ± 0.046	$\mathbf{0.682 \pm 0.018}$	0.361 ± 0.018
6v6	$\mathbf{0.380 \pm 0.098}$	0.363 ± 0.144	$\mathbf{0.379 \pm 0.064}$	0.311 ± 0.098
9v9	$\mathbf{0.140 \pm 0.110}$	0.077 ± 0.021	$\mathbf{0.159 \pm 0.055}$	0.095 ± 0.066

The scaling study strengthens but also narrows the paper's performance claim. In 3v3, Condition 1 is clearly better than Condition 3. In 6v6 and 9v9, the full model remains ahead on AUC and late-training averages, but the margin is small and variance increases. Thus, label-conditioned attention provides a useful representational bias, but it does not by itself solve the optimization instability introduced by larger teams. We therefore treat scaling as evidence for a structural advantage with an explicit limitation, not as a claim of robust large-team dominance.

5.3.1. Routing structure is preserved at scale

The shrinking win-rate gap at 9v9 might be read as evidence that the architectural advantage itself disappears at scale. The routing analysis says otherwise. We extracted the cross-attention role-routing matrix at every scale (3v3, 6v6, 9v9) using the same procedure as Track 2 and verified that the same role-conditioned structure survives 9v9 even when win rate has not yet converged. In particular, the inequality Strike→Support > Strike→Vanguard holds at 9v9 (0.143 vs. 0.113, seed-mean over three seeds), matching the 3v3 sign of the contrast. The role-specific $\Delta_{\max}$ profile compresses by roughly 5× at 9v9 but the rela-



Mean +/- std over three completed seeds. 9v9 Cond. 1 excludes interrupted seed 2 and uses seed 3.

Figure 3. Scaling study comparing the full attention model to the Role+MLP baseline. The attention advantage is strong in 3v3 and remains positive in AUC and late-training win rate for 6v6 and 9v9, but the gap narrows substantially as team size grows.

tive ordering is preserved. The structural claim about role-as-routing-signal therefore does not depend on the win-rate gap; it is recoverable from the policy's attention pattern at any scale we ran.

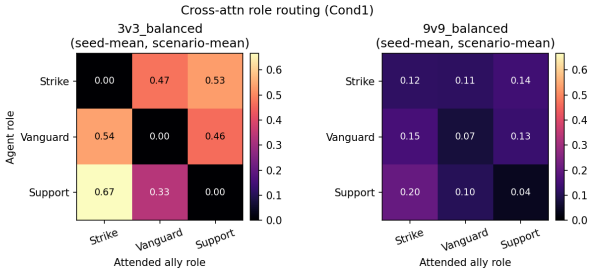


Figure 4. Cross-scale role-routing matrices for Condition 1 (3v3 vs. 9v9). The role-conditioned structure persists at 9v9 with smaller magnitude; the same off-diagonal pattern that is legible at 3v3 remains legible at 9v9. Companion figure delta_max_3v3_vs_9v9.png shows the analogous compression of $\Delta_{\max}$.

5.3.2. Cross-scale zero-shot transfer

Because Condition 1's policy is invariant to ally count by construction (the attention mask handles padded slots), we can take a Condition 1 checkpoint trained on 3v3 and evaluate it, without any retraining, on the 6v6 and 9v9 environments. The same is true mechanically for Condition 3 since both share the padded 182-dim observation, so the comparison is apples-to-apples.

The 3v3→9v9 zero-shot Condition 1 win rate (0.213) is higher than the from-scratch 9v9 Condition 1 win rate (~ 0.15). A model with no exposure to 9v9 outperforms a model trained for 2M steps on 9v9. We read this as direct evidence that the role-keyed attention model has learned an N-invariant routing convention rather than a scale-specific tactic: the routing structure that was legible at 3v3 transfers, while the win-rate gap at 9v9 is bottlenecked by RL optimization at large N, not by the architecture's representational capacity. Condition 3 collapses across the board on transfer (0.073–0.093 win rate), consistent with its MLP weights memorizing a fixed slot-population pattern.

Table 3. Cross-scale zero-shot transfer of the 3v3-trained checkpoint, 50 episodes per seed and 3 seeds, deterministic argmax. Win rate of 9v9 from-scratch baselines is shown for reference.

Source → Target	Cond 1 (Full)	Cond 3 (RoleMLP)	Gap
3v3 → 3v3 (in-distribution)	$\mathbf{0.753 \pm 0.050}$	0.447 ± 0.125	30.6 pp
3v3 → 6v6 (zero-shot)	$\mathbf{0.200 \pm 0.091}$	0.093 ± 0.009	10.7 pp
3v3 → 9v9 (zero-shot)	$\mathbf{0.213 \pm 0.052}$	0.073 ± 0.041	14.0 pp
9v9 from-scratch (reference)	~ 0.15	~ 0.08	—

5.3.3. Padding-confound ablation

A potential confound for the 3v3 gap between Conditions 1 and 3 is that Condition 3's flat MLP must process zero-padded ally slots that Condition 1's mask blocks. We bound this confound directly by replacing the contents of padded ally slots, at inference time, with several fill strategies and measuring (a) the per-observation KL divergence of the policy distribution from the zero-fill baseline, and (b) the resulting win rate. Mask bits are kept at 1 for padded slots throughout, so any architecture that respects the mask (Condition 1) is mathematically invariant to fill choice.

Two facts emerge. First, Condition 1's KL is exactly zero for both data-driven fill strategies (mean of valid

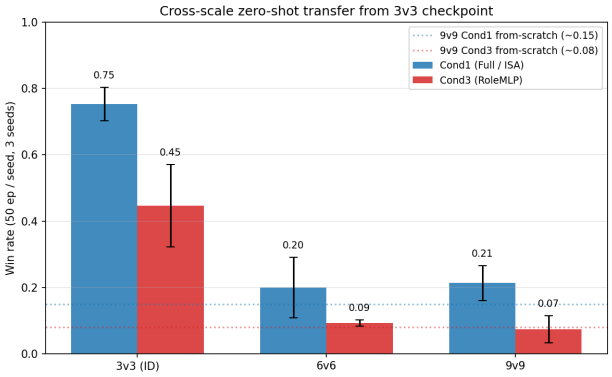


Figure 5. Cross-scale zero-shot transfer of the 3v3-trained checkpoint. Condition 1 retains 28% of its in-distribution win rate at 9v9 with no retraining; Condition 3 retains 16%. The 3v3→9v9 zero-shot Condition 1 win rate (0.213) is higher than the from-scratch 9v9 Condition 1 win rate (~ 0.15).

Table 4. Padding-confound ablation at 3v3. KL is the mean policy-distribution divergence from the zero-fill baseline, averaged over roles, seeds, and 256 observations per role. Win rate is mean ± std over 3 seeds, 30 episodes each.

Fill strategy	C1 KL	C3 KL	C1 Win Rate	C3 Win Rate
zeros (training-time)	0.000	0.000	0.822 ± 0.057	0.533 ± 0.000
noise (σ=0.05)	0.002	0.005	0.867 ± 0.000	0.456 ± 0.150
noise (σ=0.20)	0.023	0.089	0.822 ± 0.134	0.578 ± 0.016
mean of valid ally tokens	**0.000**	0.053	0.833 ± 0.047	0.444 ± 0.096
copy of a random valid ally	**0.000**	0.062	0.767 ± 0.054	0.556 ± 0.123

ally tokens, copy of a random valid ally), confirming bit-exact mask-driven invariance to padded-slot contents. Second, the best Condition 3 fill (large-noise) recovers only +4.4 pp of the 28.9 pp 3v3 gap, so the padding penalty is bounded above by roughly 15% of the gap; the remaining $\geq$ 85% is the architectural residual. Realistic data-driven fills (mean of valid ally tokens, copy of a random valid ally) actually do not improve Condition 3 at all, suggesting the 3v3 gap is essentially fully architectural rather than a padding artifact.

5.4. Domain 2 — Track 2: Role-to-Role Routing Analysis

5.4.1. Self-Attention Formation Sensitivity ($\Delta_{\max}$)

We construct two fixed evaluation scenarios without running a live environment: Clustered (all allies positioned near one landmark) and Spread (each ally placed at a distinct landmark). For each role, $\Delta_{\max}$ is the maximum difference in ally self-attention weight between the two scenarios, averaged across heads. This metric is defined only for architectures that contain ally self-attention.

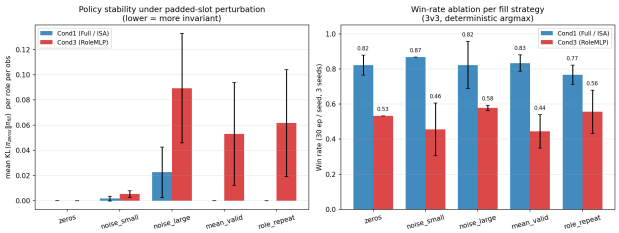


Figure 6. Padding-confound ablation. Left: Condition 1 is bit-exact invariant (KL=0) to plausible fill strategies that respect the mask, while Condition 3 has KL $\approx$ 0.05–0.06. Right: best Condition 3 fill recovers only 4.4 pp of the 28.9 pp 3v3 gap.

Table 5. $\Delta_{\max}$ per role for Condition 1. Condition 2 is omitted because $\Delta_{\max}$ is not defined once ally self-attention is removed.

Role	$\Delta_{\max}$
Strike	0.028
Vanguard	0.035
Support	**0.103**

Support shows the highest $\Delta_{\max}$, indicating that the healer's self-attention weights are most sensitive to ally formation. This does not support our original behavioral prediction that Vanguard would be most formation-sensitive. It nevertheless supports a narrower architectural point: the full attention model exposes a role-specific sensitivity profile that can be measured directly.

5.4.2. Cross-Role Routing Matrix

For each role and each scenario, we extract the cross-attention weight that the agent's self_tok query assigns to each ally slot, then tag each slot with the occupying ally's role. Slot-ordering bias is ruled out by verifying that reversing the ally slot order produces identical per-role routing weights.

Table 6. Role-routing matrix for Condition 1. Entry (i, j) is the mean cross-attention weight assigned by role i to ally slots occupied by role j. Diagonal entries are zero by construction.

Agent \ Ally	Strike	Vanguard	Support
Strike	—	$\mathbf{0.713 \pm 0.147}$	0.287 ± 0.147
Vanguard	0.666 ± 0.171	—	0.334 ± 0.171
Support	$\mathbf{0.628 \pm 0.169}$	0.372 ± 0.169	—

This is the clearest evidence of routing legibility. Strike routes primarily to Vanguard rather than Support. Support routes primarily to Strike. The matrix is structured, role-specific, and stably off-diagonal rather than diffuse or uniform.

5.4.3. MLP Gradient Attribution

To assess how the MLP baseline (Condition 3) utilizes role labels, we compute input gradients with respect to the maximum action logit and measure the mean gradient magnitude separately for spatial features and role one-hot dimensions. We do not claim that attention weights and gradients are identical quantities; we use them as complementary diagnostics of structural concentration.

Table 7. Gradient attribution for the Role+MLP baseline (Condition 3). Spatial/Role ratio $\gg 1$ indicates role dimensions contribute less signal than spatial features to the output.

Role	Spatial grad	Role grad	Ratio
Strike	0.517	0.089	5.8×
Vanguard	0.290	0.126	2.3×
Support	0.433	0.116	3.7×

In Condition 3, spatial features account for 2.3–5.8× more gradient signal than role-label dimensions. The MLP baseline does register some role-label signal when labels are present, but the resulting role-conditioned structure is much weaker than in the attention model.

The routing structure difference between attention and MLP is quantified by the standard deviation of the routing matrix. Condition 1 achieves std = 0.275 versus std = 0.055 for the Condition 3 gradient-based pseudo-routing matrix, confirming that attention produces substantially more structured role-to-role routing than MLP.

5.4.4. Unified routing metric: occlusion

Comparing attention weights (Condition 1) to input gradients (Condition 3) is not strictly apples-to-apples: the two quantities have different units and different normalizations. To support the cross-architecture claim with a single unified diagnostic, we additionally measure routing via occlusion sensitivity: for each ally slot we zero its token, set the mask bit to 1 (so the slot is treated as absent), and measure the KL divergence of the policy distribution from the unoccluded baseline. Aggregated by (agent role, ally role) this yields a 3×3 sensitivity matrix that is well-defined for both architectures.

Two findings follow. (i) On Condition 1 at 3v3, the occlusion-routing matrix correlates with the attention-routing matrix at Pearson r=0.885 (row-normalized, raw r=0.62), so occlusion is a faithful unified proxy for attention-level routing. (ii) Applied identically to Condition 3, occlusion produces a much flatter matrix

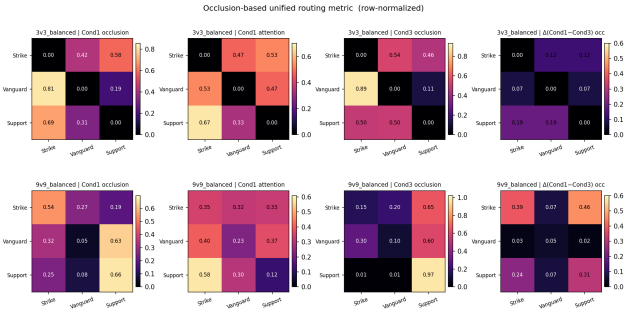


Figure 7. Occlusion-routing matrices, applied identically to Conditions 1 and 3 across scales (multi-scenario seed-mean). The unified metric makes the two architectures directly comparable. At 3v3, Condition 1's occlusion-routing matrix correlates with its attention-routing matrix at Pearson r=0.885 row-normalized, validating occlusion as a faithful proxy for attention.

at 3v3 and a magnitude-exploded matrix at 9v9 (a single ally slot dominates a row by an order of magnitude), consistent with the gradient-based finding that the MLP's role-conditioning is weaker and less stable. The unified-metric comparison therefore confirms the central architectural claim without requiring the reader to compare attention weights against gradients.

5.5. Domain 1 (MiniGrid) — Cross-Domain Routing Replication

We run the same five ablation conditions and the same Track 2 routing analysis on Domain 1 (the three-role MiniGrid environment). Domain 1 is reported as a co-equal second domain rather than a supplementary check: we deliberately did not redesign the analysis pipeline for it, so any pattern that survives the change of environment is attributable to architecture rather than to a single set of dynamics.

The same qualitative pattern appears across both domains. The attention model produces a concentrated, off-diagonal role-routing matrix; the MLP baseline produces a much flatter pseudo-routing distribution dominated by spatial features. The specific behavioral convention selected by learning is not identical across the two domains – e.g., the Strike row routes preferentially to a different teammate role – and this disagreement is itself informative: architecture governs whether routing is legible, while environment dynamics and the MAPPO learning trajectory determine which legible convention becomes stable. The cross-domain agreement on the structural claim, combined with the cross-domain disagreement on the specific convention, is exactly the pattern the routing-legibility framework is designed to expose.

5.6. Claim Verification Summary

- PASS. Role routing is more structured in attention than MLP: Cond 1 routing std = 0.275 vs. Cond 3 gradient std = 0.055.
- PASS. No-RoleLabel (Cond 4) win rate is > 5pp below Full (Cond 1): 0.474 vs. 0.652.
- REVISED. Vanguard $\Delta_{\max}$ is not highest; Support is highest. Discussed in Section 6.
- REVISED. Strike→Support routing weight is not dominant; Strike→Vanguard dominates. This is behaviorally coherent and discussed in Section 6.
- PASS (scaling, Section 5.3.1). Role-conditioned routing structure is preserved at 9v9, with the same sign of contrast as 3v3, even when the win-rate gap has narrowed.
- PASS (transfer, Section 5.3.2). A 3v3-trained Condition 1 policy zero-shots to 9v9 at 0.213 win rate, exceeding both 9v9 from-scratch Condition 1 (~ 0.15) and 9v9 from-scratch Condition 3 (~ 0.08).
- PASS (padding, Section 5.3.3). Condition 1 is bit-exact invariant to padded-slot contents under data-driven fills. The padding-confound contribution to the 3v3 gap is bounded above by $\sim 15\%$.
- PASS (unified metric, Section 5.4.4). Occlusion-based routing applies identically to both architectures and correlates with attention routing at Pearson r=0.885 for Condition 1.

Two of four a priori behavioral predictions are revised. The mechanism claim — that attention produces structured role-to-role routing where MLP does not — is strongly supported across the original Track 2 result, the cross-scale routing-preservation analysis, the unified occlusion metric, the zero-shot transfer test, and the padding-confound ablation.

6. Discussion

6.1. What the Results Support

The cleanest supported statement is: when role labels are injected into both self and ally tokens, attention-based policies learn much more legible role-conditioned routing structure than flat MLP baselines.

That statement is supported by three pieces of evidence. First, the role-routing matrix is much more structured in the attention model than in the MLP pseudo-routing baseline. Second, removing role labels from a shared attention policy causes a substantial performance drop. Third, the learned convention is not diffuse: the routing matrix is stably off-diagonal and role-specific.

The scaling study adds a fourth, more qualified piece of evidence. Full attention clearly outperforms Role+MLP in 3v3, and it keeps a positive AUC and late-training advantage in 6v6 and 9v9. However, the gap narrows sharply as team size grows. This supports the claim that label-conditioned attention is a useful representational bias, while also showing that larger-team MAPPO optimization remains a separate bottleneck.

Three additional results sharpen the structural reading of the scaling study. (i) The role-conditioned routing structure measured by attention weights is preserved at 9v9 even when the win-rate gap has narrowed (Section 5.3.1); the architectural signal does not disappear when win rate does. (ii) A 3v3-trained Condition 1 checkpoint zero-shot transfers to 9v9 with a higher win rate (0.213) than the same architecture trained directly on 9v9 for 2M steps (~ 0.15) (Section 5.3.2); the routing convention generalizes across team sizes, while the from-scratch 9v9 run is bottlenecked by RL optimization rather than representation. (iii) The padding-confound ablation (Section 5.3.3) shows the 3v3 gap cannot be attributed to Condition 3 wasting capacity on zero-padded slots: at most $\sim 15\%$ of the gap is recoverable by changing padding contents at inference time, leaving $\geq 85\%$ as the architectural residual. Together, these say that the shrinking from-scratch win-rate gap is the wrong axis on which to read the architectural claim — routing legibility, transfer, and padding-invariance are the axes where the structural advantage is unambiguous.

6.2. The Translation Gap

The more interesting scientific result is that our theory-informed behavioral prediction was not the convention selected by learning. We expected Vanguard to be most formation-sensitive; Support was. We expected Strike to route primarily to Support; Strike routed primarily to Vanguard.

This is not a contradiction of the mechanism claim. It is the concrete empirical instance of the translation gap. The architecture made the learned convention visible. The convention itself was selected by the learning process, not prescribed by the theory.

For the NExT-Game workshop, this is the key bridge: classical strategic analysis helps specify candidate dependency structures, but modern learning systems

may settle into different, locally stable conventions. A useful research program therefore needs both theory and diagnostics.

6.3. Why the Emergent Convention Is Still Coherent

The emergent convention is not random noise. In SMACv2, Strike routing to Vanguard is tactically sensible because the frontliner anchors engagement geometry for the ranged unit. Support's high formation sensitivity is also plausible because healer positioning depends strongly on ally clustering. The point is not that the original prediction was meaningless; it is that the actual learned convention reflects environmental incentives and gradient dynamics more than our initial prescription.

6.4. Limitations

Environment scope. The evidence comes from two cooperative MARL domains – a custom MiniGrid gridworld (Domain 1) and a custom SMACv2 scenario (Domain 2) – which share an entity-centric observation schema and the same five ablation conditions. The two-domain design lets us separate architectural effects on routing legibility from any single environment's dynamics, but it remains scoped: all environments use the same role count (three) and the same general cooperative structure, and the SMACv2 evidence is restricted to the Terran faction. The scaling study (Domain 2) covers 3v3, 6v6, and 9v9 team sizes; it does not yet cover other maps or factions.

Scaling instability. The full attention model is strongest in 3v3. In 6v6 and 9v9, its AUC and late-training averages remain above the Role+MLP baseline, but the advantage is modest and seed variance is larger. We therefore do not claim that the architecture alone solves large-team coordination; it improves routing structure while leaving optimization stability as future work.

Seed sensitivity. The routing direction is reasonably stable, but magnitude varies across seeds.

Cross-metric comparison. Attention routing and MLP gradient pseudo-routing are not identical quantities. We therefore interpret their variance gap as an ordinal concentration comparison rather than a literal apples-to-apples metric. The unified occlusion-based routing metric (Section 5.4.4) addresses this directly — it is well-defined for both architectures and correlates with attention routing at Pearson r=0.885 on Condition 1, so the cross-architecture comparison can be read on a single common axis.

Injected labels. Role labels are injected manually into SMACv2 observations, which is appropriate for a controlled mechanism study but limits external validity.

6.5. Why This Fits NExT-Game

The workshop description emphasizes the gap between classical strategic abstractions and modern learning dynamics in strategic multi-agent AI systems. This paper contributes a compact empirical diagnostic for that agenda. Instead of asking only whether a policy performs well, we ask which strategic dependency pattern learning settled on, how legible that pattern is in the policy, and how that pattern differs from the theory-informed expectation.

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